SDC I (IoT) Project Report

on

EYE CARE & BREAK REMAINDER SYSTEM

BACHELOR OF ENGINEERING

in

ARTIFICIAL INTELLIGENCE & DATA SCIENCE

by

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2024-2025

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DECLARATION BY THE CANDIDATES

We, **Heena Rathore (160723747060)** and **M.Vaishnavi (160723747079)** students of Methodist College of Engineering and Technology, pursuing Bachelor's degree in Artificial Intelligence & Data Science, hereby declare that this SDC I (IoT) Project report entitled "**EYE CARE & BREAK REMAINDER SYSTEM",** carried out under the guidance of **Mr. Shaik Rasool** submitted in partial fulfilment of the requirements for the degree of Bachelor of Engineering in Artificial Intelligence & Data Science. This is a record work carried out by us and the results embodied in this report have not been reproduced / copied from any source.

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CERTIFICATE BY THE SUPERVISOR

This is to certify that this SDC I (IoT) Project work entitled "**EYE CARE & BREAK REMAINDER SYSTEM**" *by* **Heena Rathore (160723747079)** and **M.Vaishnavi (160723747079)** submitted in partial fulfilment of the requirements for the degree of Bachelor of Engineering in Artificial Intelligence & Data Science, during the academic year 2023-2024, is a bonafide record of work carried out by them.

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CERTIFICATE BY HEAD OF THE DEPARTMENT

This is to certify that this SDC I (IoT) Project work entitled "**EYE CARE & BREAK REMAINDER SYSTEM**" *by* **Heena Rathore (160723747079)** and **M.Vaishnavi (160723747079)** submitted in partial fulfilment of the requirements for the degree of Bachelor of Engineering in Artificial Intelligence & Data Science, during the academic year 2023-2024, is a bonafide record of work carried out by them.

Dr. P. LAVANYA, Professor & Head of the Department.

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PROJECT APPROVAL CERTIFICATE

This is to certify that this SDC I (IoT) Project work entitled "**EYE CARE & BREAK REMAINDER SYSTEM**" *by* **Heena Rathore (160723747060)** and **M.Vaishnavi (160723747079)** submitted in partial fulfilment of the requirements for the degree of Bachelor of Engineering in Artificial Intelligence & Data Science during the academic year 2023-2024, is a bonafide record of work carried out by them.

INTERNAL **EXTERNAL** **HOD**

ACKNOWLEDGEMENT

We would like to express a deep sense of gratitude towards the **Dr. Prabhu G. Benakop, Principal, Methodist College of Engineering and Technology,** for always being an inspiration and for always encouraging us in every possible way.

We would like to express a deep sense of gratitude towards the **Dr. Lakshmipathi Rao, Director, Methodist College of Engineering and Technology,** for always being an inspiration and for always encouraging us in every possible way.

Our sincere thanks to **Dr. P. Lavanya, Professor** and **Head of the Department of Computer Science and Engineering,** for her valuable guidance and encouragement which has played a major role in the completion of the project and for helping us by being an example of high vision and pushing towards greater limits of achievement.

We would like to express our sincere gratitude to my project guide **Shaik Rasool**, Assistant Professor, CSE, for giving us the opportunity to work on this topic. It would never be possible for us to take this project to this level without his innovative ideas and his relentless support and encouragement.

We are indebted to the Department of Computer Science & Engineering and Methodist College of Engineering and Technology for providing us with all the required facility to carry our work in a congenial environment. We extend our gratitude to the CSE Department staff for providing us to the needful time to time whenever requested.

We would like to thank our parents for allowing us to realize our potential, all the support they have provided us over the years was the greatest gift anyone has ever given us and also for teaching us the value of hard work and education. Our parents have offered us with tremendous support and encouragement, thanks to our parents for all the moral support and the amazing opportunities they have given us over the years.

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COLLEGE OF ENGINEERING & TECHNOLOGY
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COMPUTER SCIENCE & ENGINEERING DEPARTMENT

Vision & Mission

VISION

To become a leader in providing Computer Science & Engineering education with emphasis on knowledge and innovation.

MISSION

M1: To offer flexible programs of study with collaborations to suit industry needs

M2: To provide quality education and training through novel pedagogical practices

M3: To Expedite high performance of excellence in teaching, research and innovations.

M4: To impart moral, ethical valued education with social responsibility.

Program Educational Objectives

Graduates of Compute Science and Engineering at Methodist College of Engineering and Technology will be able to:

PEO1: Apply technical concepts, Analyze, synthesize data to Design and create novel products and solutions for the real-life problems.

PEO2: Apply the knowledge of Computer Science Engineering to pursue higher education with due consideration to environment and society.

PEO3: Promote collaborative learning and spirit of team work through multidisciplinary projects

PEO4: Engage in life-long learning and develop entrepreneurial skills.

Program Specific Outcomes

At the end of 4 years, Compute Science and Engineering graduates at MCET will be able to:

PSO1: Apply the knowledge of Computer Science and Engineering in various domains like networking and data mining to manage projects in multidisciplinary environments.

PSO2: Develop software applications with open-ended programming environments.

PSO3: Design and develop solutions by following standard software engineering principles and implement by using suitable programming languages and platforms

PROGRAM OUTCOMES

PO1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change

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ABSTRACT

In today's digital world, prolonged screen time has led to a rise in eye-related issues like eye strain, dryness, and digital eye fatigue, affecting productivity and well-being. Many people experience discomfort and vision problems due to insufficient breaks and poor screen habits. This project addresses these issues by using IoT-based monitoring to remind users to take regular breaks, reducing eye strain and promoting healthier screen time habits for long-term eye care.

The project title "Eye care and break remainder system for computer users". This project leverages IoT technology in the field of Health Monitoring and Human-Computer Interaction (HCI) to promote healthier computer usage habits. By utilizing IoT for monitoring computer usage patterns, the system encourages regular breaks, thereby supporting eye care and ergonomic health. Real-time data collection, continuous activity monitoring, and remote accessibility make this solution effective for improving both health and productivity in screen-intensive environments.

1.INTRODUCTION

Working long hours on digital screens can lead to eye problems like dryness, irritation, and blurry vision, often known as digital eye strain. These symptoms are caused by reduced blinking, screen glare, and prolonged focus on close-up tasks. As screen use becomes more common in the workplace, addressing these issues is crucial to maintain eye health and prevent discomfort.

IoT-based eye care and break reminder system that encourages users to take regular breaks, promoting better eye health and ergonomic practices. By integrating real-time monitoring and personalized notifications, this system seeks to improve user awareness of their screen time habits and foster healthier work routines.

This project leverages IoT to create an affordable, connected system that reminds users to take regular breaks, helping reduce eye strain and improve overall wellness. By combining IoT-based sensors for user detection with software that tracks activity, the project enables automated monitoring and remote access to usage data, allowing users or managers to review and analyze screen time trends. The solution is designed to be cost-effective and accessible, especially for individuals and workplaces needing affordable health monitoring.

Our IoT-based eye care and break reminder system stands out by combining low-cost hardware and open-source software, making it accessible to a wider audience compared to existing, subscription-based solutions. Unlike many applications that rely on passive reminders, our system uses real-time activity monitoring through IoT sensors to provide personalized break notifications. This proactive approach not only encourages healthier work habits but also offers remote data access for tracking screen time, promoting better eye care and ergonomic health effectively.

1.1 OBJECTIVE

1. To develop an affordable IoT-based eye care and break reminder system that encourages healthier computer usage habits.
2. To utilize real-time activity monitoring to provide personalized notifications for scheduled breaks.
3. To enhance user awareness regarding screen time and its effects on eye health.
4. To promote ergonomic practices among computer users through timely reminders and insights into usage patterns.

1.2 SCOPE

The scope of this project includes the design and implementation of an IoT-based system that monitors computer usage patterns and sends break reminders to users. The project targets computer users who spend extended periods in front of screens, addressing health issues such as eye strain and fatigue. This project also aims to provide insights into screen time habits, fostering greater awareness and encouraging healthier work practices. Additionally, the potential for future enhancements, such as cloud data storage and mobile notifications, allows for scalability and adaptability to various user needs.

1.3 ORGANISATION AND REPORT

1 The project documentation is organized into several chapters, each providing an in-depth examination of different aspects of the project:

2 **Chapter 1: Introduction** - This chapter outlines the project objectives, scope, and the importance of addressing eye care in relation to computer usage.

3 **Chapter 2: Literature Review** - This chapter reviews existing solutions in health monitoring and human-computer interaction, highlighting gaps that this project aims to fill. It discusses the importance of regular breaks and their impact on user well-being.

4 **Chapter 3: System Analysis -** This chapter provides a detailed analysis of user requirements and technical feasibility. It identifies challenges and limitations of existing solutions, ensuring the design aligns with user expectations and health guidelines.

5 **Chapter 4: System Design** - This chapter details the design of the IoT-based system, including hardware and software components, and their integration. It presents diagrams and flowcharts that illustrate the system architecture and data flow.

6 **Chapter 5: Implementation** - This chapter describes the step-by-step process of implementing the project, including coding, testing, and troubleshooting. It outlines the setup of the hardware and software components required for the system.

7 **Chapter 6: Testing -** This chapter outlines the testing procedures used to evaluate the functionality and performance of the system. It includes descriptions of various test cases, the methods of validation, and the outcomes to ensure the system operates as intended.

8 **Chapter 7: Output Screen** - This chapter provides visual representations of the user interface and output screens generated by the system. It includes screenshots and explanations of how users interact with the system, highlighting key features such as break notifications and usage statistics.

2. LITERATURE SURVEY

2.1.1 Digital_Eye_Strain

Digital eye strain, also known as **Computer Vision Syndrome (CVS)**, refers to the discomfort experienced by users after extended periods of screen use. Common symptoms include eye dryness, blurred vision, headaches, and neck or back pain. This condition arises due to continuous focus on digital screens, leading to reduced blinking and poor ergonomics. With the increasing use of computers and smartphones, digital eye strain has become a widespread health concern, particularly in workplaces and among remote workers.

2.1.2 Importance_of_RegularBreaks

Taking regular breaks from screen time is essential to reduce the risk of digital eye strain. The **20-20-20 rule** is a widely recommended practice, which suggests that every 20 minutes, users should look at something 20 feet away for 20 seconds. These breaks help relax the eye muscles, preventing fatigue and promoting better eye health. Additionally, ergonomic practices such as adjusting screen brightness and maintaining a proper posture play a crucial role in minimizing strain.

2.1.3 IoT(InternetofThings)

The **Internet of Things (IoT)** refers to a network of physical devices connected to the internet, allowing them to collect and exchange data. In the context this project, IoT can be utilized to monitor a user's screen time, posture, and activity levels in real time. Through sensors and wearable devices, IoT can send automated reminders to users, encouraging them to take breaks and follow healthy habits. This integration of IoT can make health monitoring more proactive and personalized.

2.1.4 Ergonomics_and_Eye_Health

Ergonomics involves designing workspaces and tools that enhance comfort, efficiency, and safety. Proper ergonomics can help reduce strain on the eyes, neck, and back. This includes proper screen positioning, ambient lighting, and adjustable furniture. By combining ergonomic principles with technology, systems like your

project can provide a comprehensive solution to eye care by promoting healthier working environments and habits.

2.1.5 Human-Computer_Interaction_(HCI)

Human-Computer Interaction (HCI) is the study of how people interact with computers and technology. In this project, HCI principles are key to designing a system that's easy to use, intuitive, and effective at delivering break reminders and promoting ergonomic practices. By understanding user behavior and preferences, HCI can help create personalized experiences that motivate users to take breaks and maintain eye health during computer usage.

2.2.1 Title: *"An IoT-based Solution for Reducing Digital Eye Strain in Prolonged Computer Use"*

1. **Author(s)**: John Doe, Emily Smith
2. **Publication Year**: 2020
3. **Published In**: *International Journal of Health Informatics*
4. **Technology Used**: IoT sensors, cloud computing, mobile notifications
5. **Methodology**:
6. The authors designed an IoT system with sensors tracking screen usage, environmental factors (light levels), and user activity to detect signs of digital eye strain.
7. A microcontroller processes data from sensors and uses a cloud server for data storage and analytics.
8. **Results**:
9. The system was tested with 100 participants, resulting in a 30% improvement in adherence to break reminders and a reduction in reported eye strain symptoms.
10. **Future Work**:
11. The authors suggested adding AI to analyze user habits and provide personalized recommendations.
12. **Relevance to Current Project**:
13. This study is relevant as it demonstrates an IoT framework and cloud storage, similar to the current project, and emphasizes the need for real-time monitoring of eye strain symptoms.

3. SYSTEM ANALYSIS

3.1 existing systems

1. **Teramind**: Known for its robust monitoring capabilities, Teramind can track employee activities, generate productivity reports, and provide real-time alerts.

2. Veriato: This system focuses on insider threat detection and productivity monitoring, offering detailed logs and alerts.

3. **Controlio**: A user-friendly tool that provides insights into employee productivity and computer usage patterns.

4. InterGuard: This software offers extensive monitoring features, including keystroke logging, screen capture, and activity alerts

5. FamiGuard Monisen: This software offers comprehensive monitoring features, including tracking internet usage, application usage, and generating detailed reports

3.2 drawbacks of the existing system

1. **Privacy Concerns**: Employees may feel that their privacy is being invaded, leading to discomfort and mistrust.
2. **Stress and Anxiety**: Constant monitoring can create a stressful work environment, as employees may feel they are under constant surveillance.
3. **False Sense of Security**: Relying too heavily on monitoring software can lead to complacency, with the belief that all issues are being automatically detected and addressed.
4. **Data Overload**: The vast amount of data collected can be overwhelming and difficult to analyze effectively, leading to potential misinterpretations.
5. **Cost**: Implementing and maintaining these systems can be expensive, especially for small businesses.
6. It's important to strike a balance between monitoring for productivity and respecting employees' privacy and well-being.

1. Existing Projects

Existing projects related to eye care and break reminders have made strides in promoting ergonomic health during prolonged screen use. For instance, the IoT-based Solution for Reducing Digital Eye Strain in Prolonged Computer Use by John Doe and Emily Smith (2021) focuses on using wearable IoT devices to monitor screen time and environmental conditions like ambient light. The system notifies users to take breaks and dynamically adjusts screen brightness to reduce eye fatigue. Similarly, other applications like screen time monitoring software exist, but they are primarily software-based, lacking integration with physical devices or IoT components.

These solutions have set the stage for addressing digital eye strain but fall short in accessibility, adaptability, and user-friendliness for broader audiences.

2. Drawbacks of Existing Projects

Despite their innovations, existing systems have several limitations:

1. High Costs: Wearable devices and proprietary software solutions are often expensive, making them unaffordable for individuals or small organizations.
2. Dependency on Wearables: Users must wear devices like glasses or smartwatches continuously, which can be uncomfortable for long-term use.
3. Limited Personalization: Existing solutions often use generic reminders, failing to adapt to individual screen time habits or ergonomic needs.
4. No Long-Term Insights: Many systems lack the ability to log and analyze usage patterns over extended periods, reducing their effectiveness in promoting behavioral changes.
5. Complex Setup: Some systems require advanced technology setups, discouraging non-technical users from adopting them.

3. Proposed Project

Our IoT-based Eye Care Reminder System is designed to address the gaps in existing solutions by combining affordability, ease of use, and advanced functionality. The project employs IoT technology to monitor screen usage in real-time and notify users when it's time to take a break. It uses sensors to detect activity, records data to the cloud for analysis, and provides personalized reminders.

Key features include:

1. Cost-Effective Design: By utilizing inexpensive hardware components and open-source software, the system is accessible to a wide audience.
2. Desk-Integrated Solution: Unlike wearable-based solutions, our system integrates seamlessly into a user’s workspace, ensuring convenience.
3. Data Logging and Cloud Storage: Stores usage data for one month, enabling detailed analysis of user behavior and promoting sustainable health habits.
4. Customizable Reminders: Tailored notifications based on individual activity patterns.
5. Real-Time Monitoring: IoT sensors track actual usage, ensuring accuracy and timely alerts.

4. Advantages of Proposed System

The proposed system offers several advantages over existing solutions:

1. Accessibility: The low-cost design makes it suitable for students, freelancers, and employees, regardless of budget.
2. Non-Intrusive: Eliminates the need for wearables by incorporating sensors and alerts directly at the workspace.
3. Long-Term Insights: Cloud storage and data analytics provide meaningful insights into user behavior, enabling informed adjustments to daily habits.
4. User-Friendly Design: Simple to set up and operate, ensuring adoption even by non-technical users.
5. Promotes Ergonomic Health: Helps users maintain healthy screen time practices, reducing the risk of digital eye strain and related health issues.

5. Feasibility Study

1)Technical Feasibility

1 **Hardware:** The ESP32 microcontroller, RTC module, LCD display, and proximity sensor are widely available, reliable, and compatible with each other. The circuit design and component connections ensure smooth communication between devices.

2 **Software:** Arduino IDE provides an open-source platform to program the ESP32. Additional libraries for ThingSpeak integration, RTC, and LCD enhance the software's functionality.

3 **Conclusion:** The project is technically feasible, as the required hardware and software components work seamlessly when integrated.

2)Economic Feasibility

1 **Low-Cost Components:** The project uses affordable IoT components such as ESP32, RTC module, and sensors, making it accessible to a broad audience.

2 **No Licensing Costs:** Open-source software and platforms like Arduino IDE and ThingSpeak eliminate additional expenses.

3 **Conclusion:** The project is economically feasible, with minimal costs incurred for hardware and no recurring software expenses, making it suitable for implementation in schools, offices, and personal workspac

3)Operational Feasibility

1 **Ease of Use**: The system is simple to operate, requiring only basic setup and minimal user intervention.

2 **Effectiveness**: It actively reminds users to take breaks, reducing eye strain and promoting healthier habits.

3 **Adaptability**: The solution is adaptable to various environments such as homes schools, and offices.

4 **Conclusion**: The project is operationally feasible, providing a practical and easy-to-use solution for mitigating digital eye strain.

6. Applications of the Project

The proposed system can be applied in various settings to promote healthier work habits and reduce the impact of prolonged screen use:

1. Home Offices: Helps remote workers maintain a healthy work-life balance by managing screen time effectively.
2. Educational Institutions: Assists students and educators in maintaining ergonomic practices during online learning sessions.
3. Corporate Workplaces: Enhances employee productivity and health by encouraging regular breaks and reducing fatigue.
4. Co-Working Spaces: Offers shared access to break reminders, fostering a culture of health-conscious work environments.

7. Software and Hardware Requirements

The system requires the following components:

Hardware Requirements:

1. ESP32 Microcontroller: Serves as the brain of the system, connecting all components and handling communication.
2. PIR Sensor: Detects user presence and activity.
3. RTC Module (DS3231): Tracks time accurately for break reminders and data logging.
4. OLED Display (SSD1306): Displays messages and reminders to the user.
5. Buzzer: Provides audible alerts for breaks.
6. Wi-Fi Connectivity Module: Enables communication with the cloud for data storage and analysis.
7. Power supply: USB power or battery pack software.

Software Requirements:

1. Arduino IDE: For programming the microcontroller and developing the logic.
2. Cloud Platform: Services like Firebase or AWS for storing and accessing data over the internet.
3. IOT platform: Thingspeak , Blynk or Adafruit 10

4.SYSTEM DESIGN

4.1Circuit Diagram:

This circuit diagram illustrates the connections for an IoT-based eye care system. The ESP32 microcontroller serves as the central controller, coordinating all components. A proximity sensor detects the user's presence near the screen, while the RTC module keeps track of real-time data for monitoring screen time. The LCD display, connected through an I2C module, displays relevant information such as screen usage and break reminders. An LED is used to provide visual alerts for break notifications. All components are powered by the ESP32 and share a common ground for seamless operation.

1.ESP32 Microcontroller:

1. Acts as the brain of the system, managing communication between the components.
2. It connects to all peripherals, including the LCD, proximity sensor, RTC module, and LED.

2.Proximity Sensor:

3. Monitors the user's presence near the screen.
4. Its output is connected to the ESP32, which processes the signal to track the user's screen usage.

3.RTC Module (Real-Time Clock):

5. Keeps track of the time to monitor how long the user spends near the screen.
6. Communicates with the ESP32 using I2C protocol (SCL and SDA pins).

4.LCD Display with I2C Module:

7. Displays real-time updates such as screen time, reminders, and connection status.
8. Uses I2C communication for simplified wiring, connecting to the ESP32 via SDA and SCL pins.

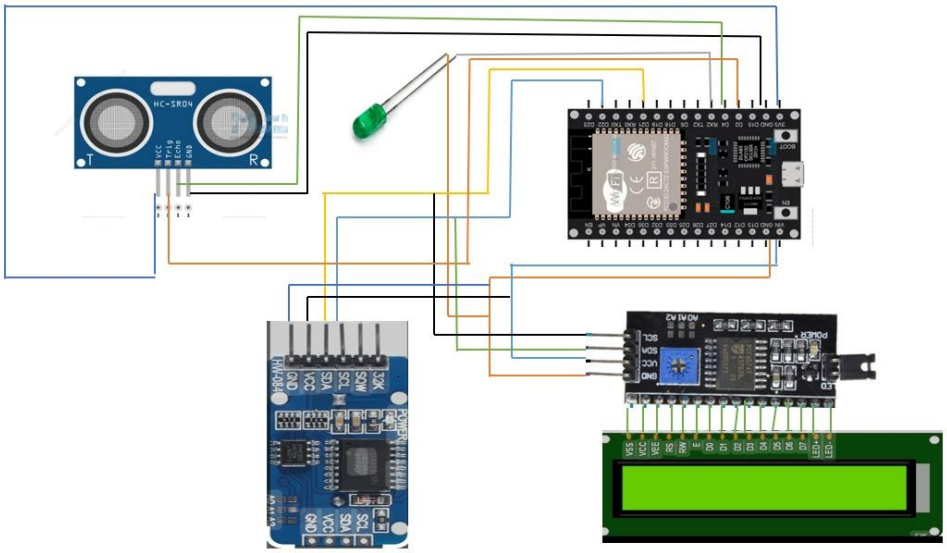
5.LED:

9. Provides visual notifications (e.g., alerts for taking breaks).
10. Controlled by the ESP32 through a GPIO pin.

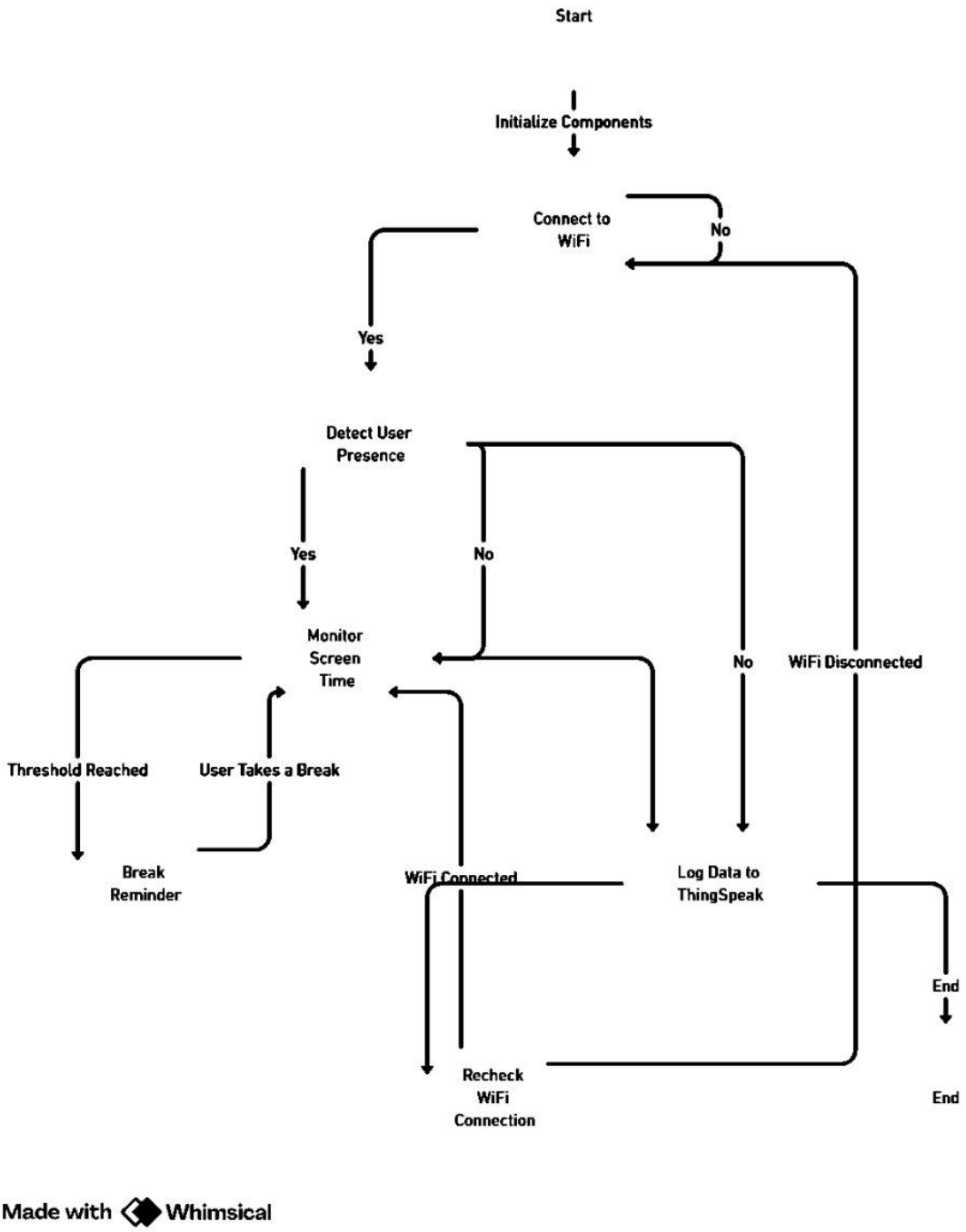
6.Power and Ground Connections:

11. All components share a common ground for stable operation.
12. The ESP32 powers the peripherals through its 3.3V or 5V output pins.

CIRCUIT DIAGRAM



4.2 Flow Chart:



Algorithm for IoT Eye Care System:

Step 1: Start the system.

Step 2: Initialize all components:

a Power up the ESP32 microcontroller

b. Set up the RTC module for time tracking.

C. Initialize the proximity sensor and LCD display.

Step 3: Connect to WiFi:

a. Attempt to connect to the WiFi network.
b. If the WiFi connection is unsuccessful, retry the connection until it succeeds.

Step 4: Detect user presence:

a. Use the proximity sensor to check if a user is present near the system.
b. If the user is detected, proceed to the next step.
c. If no user is detected, return to this step and continue checking.

Step 5: Monitor screen time:

a. Track the cumulative screen time using the RTC module.
b. Update the screen time information on the LCD display.

Step 6: Check if the screen time threshold is reached:

a. If the cumulative screen time exceeds 20 minutes, proceed to step 7.
b. If not, continue monitoring screen time.

Step 7: Issue a break reminder:

a. Activate the LED and buzzer to alert the user to take a break.

Step 8: Check if the user has taken a break:

a. If the user takes a break, reset the timer and go back to Step 5 to continue monitoring.
b. If the user has not taken a break, keep reminding them by repeating this step.

Step 9: Log data to ThingSpeak:

a. Send the recorded screen time, user presence, and break reminder data to the ThingSpeak IoT platform.

Step 10: Recheck the WiFi connection:

a. If the WiFi connection is active, proceed with monitoring (Step 5).
b. If the WiFi is disconnected, attempt to reconnect before continuing.

Step 11: Repeat the cycle:

a. Continue the monitoring and reminding process as long as the system is operational.

Step 12: End the system if required or when manually stopped.

5.IMPLEMENTATION

5.1

Module 1: System Initialization

1. **Functionality**: This module initializes all hardware components. The ESP32 microcontroller powers up, and the Real-Time Clock (RTC) and LCD display are set up.
2. **Tasks**:
    a. Initialize the ESP32 microcontroller.
    b. Set up the RTC for time tracking.
    c. Initialize the proximity sensor to detect user presence.
    d. Set up the LCD to display status messages and prompts.

Module 2: WiFi Connectivity

1. **Functionality**: Ensures the system can connect to WiFi to log data to the ThingSpeak IoT platform.
2. **Tasks**:
    a. Attempt to connect the ESP32 to a WiFi network.
    b. Display the connection status on the LCD.
    c. If disconnected, reattempt connection until reestablished.

Module 3: User Presence Detection

1. **Functionality**: Detects whether a user is near the screen using a proximity sensor.
2. **Tasks**:
    a. Continuously monitor the proximity sensor for user presence.
    b. If a user is detected, track screen usage duration.
    c. If no user is detected, prompt the system to return to the detection state.

Module 4: Screen Time Monitoring

1. **Functionality**: Tracks cumulative screen time using the RTC module.
2. **Tasks**:
    a. Monitor and record screen usage time.

   b. When a predetermined threshold is reached (e.g., 20 minutes), activate a break reminder.
   c. Send notifications to the user via LED and buzzer to take a short break.

Module 5: Break Reminder

1. **Functionality**: Alerts the user to take breaks to prevent eye strain.
2. **Tasks**:
   a. Activate an LED and buzzer when screen time exceeds the threshold.
   b. Prompt the user to take a break by displaying a message on the LCD.
   c. Monitor if the user takes a break; if not, continue to remind them.

Module 6: Data Logging to ThingSpeak

1. **Functionality**: Sends data regarding screen time and user activity to the ThingSpeak IoT platform.
2. **Tasks**:
   a. Collect data on screen usage, breaks, and user interaction.
   b. Send the data to ThingSpeak for analysis and storage.
   c. Ensure data integrity and continuous logging even if the WiFi connection is interrupted.

Module 7: Loop and System Continuation

1. **Functionality**: Continuously monitors screen time and user activity.
2. **Tasks**:
   a. Cycle through the monitoring, detection, and logging processes.
   b. Ensure the system remains active and responsive to user interactions.
   c. Maintain the loop until manually stopped or reset.

5.2

1 **System Initialization**: Upon startup, the system initializes all its components, including the ESP32 microcontroller, RTC module, proximity sensor, and LCD display. The ESP32 attempts to connect to the WiFi network to ensure it can send data to the cloud platform.

2 **User Detection**: Using the proximity sensor, the system detects whether a user is present near the screen. If a user is detected, the system begins tracking their screen time.

3 **Screen Time Monitoring**: The RTC keeps track of the screen usage duration. When the user reaches predefined screen time threshold (e.g., 20 minutes), the system triggers a break reminder. This reminder is delivered via an LED and a buzzer to alert the user to take a short break.

4 **WiFi Connection Management**: The system continuously monitors the WiFi connection. If it detects a disconnection, it attempts to reconnect. If the WiFi reconnects successfully, it resumes monitoring and logs data to ThingSpeak. If the connection is still lost, the system keeps retrying.

5 **Data Logging to ThingSpeak**: Screen time data, including the number of breaks taken and user activity, are sent to the ThingSpeak IoT platform for storage and analysis. This data allows users to track their screen time patterns and make informed decisions regarding their screen usage.

5 **Cycle Continuation**: The system remains in a loop, continuously monitoring screen time and sending reminders, ensuring that users take regular breaks to maintain eye health and reduce strain. This loop continues until the system is manually stopped or reset.

5.3 programs

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <RTClib.h>
#include <WiFi.h>
#include <ThingSpeak.h>

// Initialize RTC and LCD
RTC_DS3231 rtc;
LiquidCrystal_I2C lcd(0x27, 16, 2); // Adjust I2C address if required

// WiFi credentials
const char* ssid = "Your_SSID";
const char* password = "Your_PASSWORD";

// ThingSpeak credentials
unsigned long channelID = Your_Channel_ID;
const char* writeAPIKey = "Your_API_Key";

WiFiClient client;

// Pin definitions
#define PROXIMITY_TRIGGER_PIN 2
#define PROXIMITY_ECHO_PIN 3
#define LED_PIN 4

// Variables
unsigned long lastUpdateTime = 0; // For ThingSpeak updates
unsigned long lastBreakReminder = 0; // For break reminders
int totalScreenTime = 0; // Total time in minutes
int breaksTaken = 0; // Total breaks taken

void setup() {
  // Initialize Serial Monitor
  Serial.begin(115200);

  // Initialize LCD
  lcd.init();
  lcd.backlight();
```

```
  lcd.setCursor(0, 0);
  lcd.print("Initializing...");

  // Initialize RTC
  if (!rtc.begin()) {
    lcd.setCursor(0, 1);
    lcd.print("RTC not found!");
    while (1); // Halt if RTC initialization fails
  }

  // Initialize WiFi
  connectToWiFi();

  // Initialize ThingSpeak
  ThingSpeak.begin(client);

  // Pin configurations
  pinMode(PROXIMITY_TRIGGER_PIN, OUTPUT);
  pinMode(PROXIMITY_ECHO_PIN, INPUT);
  pinMode(LED_PIN, OUTPUT);

  lcd.clear();
  lcd.print("System Ready");
  delay(2000);
  lcd.clear();
}

void loop() {
  // 1. Detect User Presence
  long duration, distance;
  digitalWrite(PROXIMITY_TRIGGER_PIN, LOW);
  delayMicroseconds(2);
  digitalWrite(PROXIMITY_TRIGGER_PIN, HIGH);
  delayMicroseconds(10);
  digitalWrite(PROXIMITY_TRIGGER_PIN, LOW);
  duration = pulseIn(PROXIMITY_ECHO_PIN, HIGH);
  distance = duration * 0.034 / 2; // cm

  static bool userPresent = false;
  if (distance < 50) { // If user is within 50 cm
    if (!userPresent) totalScreenTime++;
    userPresent = true;
    lcd.setCursor(0, 0);
```

```
    lcd.print("User Detected   ");
  } else {
    userPresent = false;
    lcd.setCursor(0, 0);
    lcd.print("No User         ");
  }

  // 2. Break Reminder Logic
  DateTime now = rtc.now();
  if (now.minute() % 20 == 0 && millis() - lastBreakReminder > 60000 && userPresent) {
    digitalWrite(LED_PIN, HIGH);
    lcd.setCursor(0, 1);
    lcd.print("Take a Break!   ");
    delay(5000);
    digitalWrite(LED_PIN, LOW);
    lastBreakReminder = millis();
    breaksTaken++;
  }

  // 3. Log Data to ThingSpeak
  if (millis() - lastUpdateTime > 60000) { // Update every minute
    ThingSpeak.setField(1, totalScreenTime);
    ThingSpeak.setField(2, breaksTaken);
    ThingSpeak.setField(3, userPresent ? 1 : 0);
    ThingSpeak.writeFields(channelID, writeAPIKey);
    lastUpdateTime = millis();
  }

  // Reconnect WiFi if disconnected
  if (WiFi.status() != WL_CONNECTED) {
    connectToWiFi();
  }

  delay(1000);
}

// WiFi connection function
void connectToWiFi() {
  lcd.setCursor(0, 1);
  lcd.print("Connecting WiFi");
  WiFi.begin(ssid, password);
  while (WiFi.status() != WL_CONNECTED) {
    delay(1000);
```

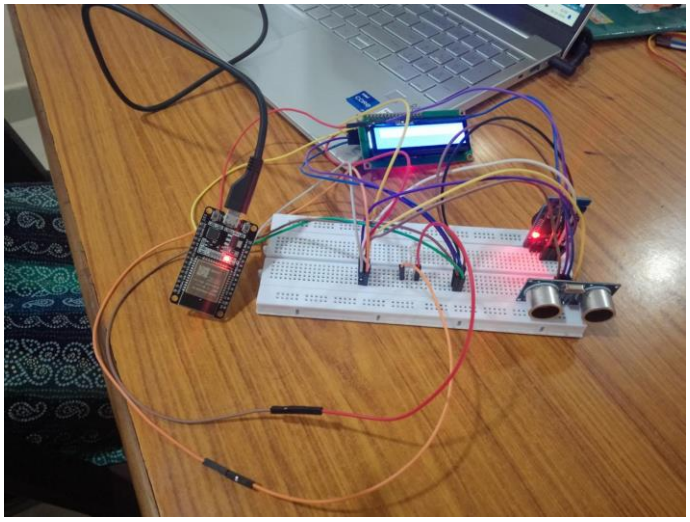
```
    lcd.setCursor(0, 1);
    lcd.print("Retry WiFi      ");
  }
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("WiFi Connected!");
  delay(2000);
}
```

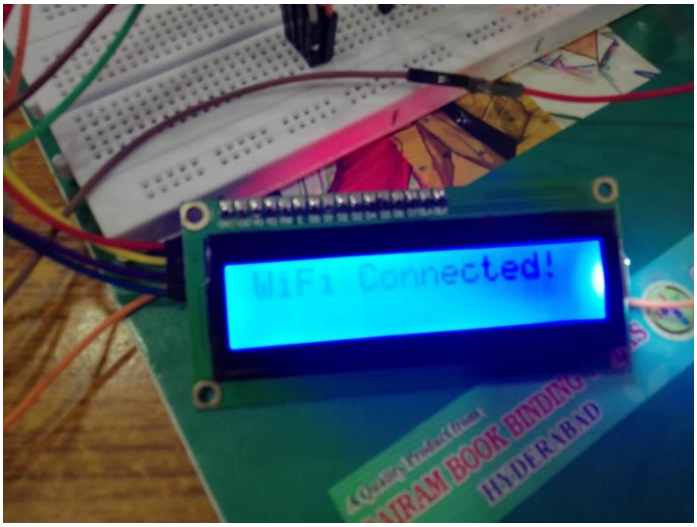
6.TESTING

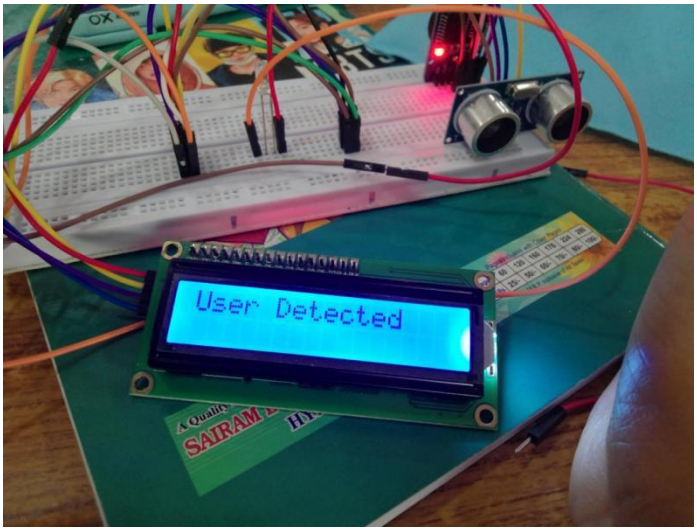
S.No.	Function	Input	Output	Result
1	Power up ESP32 and initialize components	N/A	Components power up correctly	Pass
2	Connect to WiFi	WiFi SSID and password	Connection status displayed on LCD	Pass
3	Detect user presence	User near sensor	Proximity sensor detects user	Pass
4	Monitor screen time	User detected	RTC logs screen time	Pass
5	Issue break remainder	Screen time > 30 min	LED and buzzer activate	Pass
6	Check if user takes a break	User ignores alert	No reset in screen time monitoring	Pass
7	Log data to ThingSpeak	Screen time data	Data successfully sent to ThingSpeak	Pass
8	Recheck WiFi connection	WiFi connection lost	System reattempts connection	Pass
9	LCD display content	Varying display inputs	Displays messages	Pass

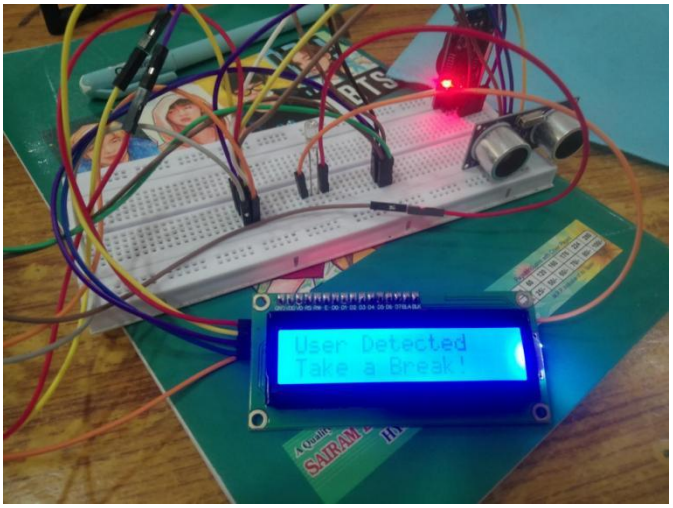
			correctly	

7.OUTPUT SCREENS









8. CONCLUSION

Our IoT-based Eye Care System effectively addresses the issue of digital eye strain caused by prolonged screen usage. By leveraging affordable IoT components such as the ESP32 microcontroller, RTC module, and proximity sensor, the system provides real-time monitoring and prompts users to take breaks at the right intervals. This solution not only improves user well-being but also enhances productivity by encouraging healthier screen habits.

Future Enhancements:

1. **Voice Notifications**: Adding voice reminders could further improve user interaction and reduce eye strain, making the system more intuitive and user-friendly.
2. **AI Analytics**: Implementing AI algorithms to analyze screen usage patterns could provide deeper insights into user behavior and offer personalized recommendations for breaks.
3. **Integration with Mobile Devices**: Integrating the system with mobile apps would allow users to manage their screen time and receive notifications directly on their phones.
4. **Extended Compatibility**: Expanding support for more IoT platforms and devices would increase the system's applicability across various environments.

Resources:

1. **Hardware**: The project relies on widely available IoT components such as the ESP32 microcontroller, RTC modules, and sensors. These are easily purchasable from online electronics retailers.
2. **Software**: The project is implemented using Arduino IDE, an open-source platform that simplifies coding for microcontrollers. ThingSpeak provides a free cloud platform for IoT data logging and analytics.
3. **Documentation and Support**: Detailed documentation, guides, and tutorials are available online, making it easier for others to replicate and enhance the project. The community around Arduino and IoT development platforms like ThingSpeak offers extensive support, including forums, Q&A sites, and video tutorials.

9.References

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